

CubeSat Simulation Code Structure

This document provides a description of the MATLAB code structure extracted from the PDF file, which simulates the orbit propagation, deorbit, and burn-up of a CubeSat.

1. Parameters

This section defines the basic variables that set up the simulation environment and CubeSat properties. These variables include:

- `R_E` : Earth radius in kilometers.
- `h_orbit` : Initial orbital altitude in kilometers.
- `r_orbit` : Initial orbital radius.
- `incl` : Inclination in degrees.
- `num_orbits` : Number of orbits to simulate.
- `total_steps` : Total simulation steps.
- `earth_rot_speed` : Earth rotation speed factor.
- `deorbit_start` : The step at which deorbit begins.
- `burn_start` : The step at which burn-up begins.

2. Orbit Propagation (with Deorbit Effect)

This section describes how the CubeSat's position is calculated at each simulation step, considering the deorbit effect. This section includes:

- Calculation of the `theta` angle using `linspace`.
- A `for` loop for each simulation step.
- Adjustment of `altitude` based on the deorbit and burn-up phases.
- Calculation of the local position (`r_local`) in the orbital plane.
- Application of the inclination rotation matrix (`R_incl`) to obtain the final 3D position (`r_sat`).

3. Figure Setup

This section sets up the plotting environment to display the simulation. This includes:

- Creating a new figure window and setting its properties.
- Labeling the axes (`xlabel` , `ylabel` , `zlabel`) and setting the figure title (`title`).

- Defining the axis limits (`axis`).

4. Earth Visualization

In this section, a 3D model of the Earth is created and visualized:

- Creating a sphere using `sphere` .
- Loading terrain data (`topo.mat`) to apply as a texture map to the Earth.
- Creating an `hgtransform` object for the Earth group.
- Plotting the Earth's surface using `surf` and applying the colormap (`colormap`).
- Setting up lighting (`camlight` , `lighting`).

5. Satellite Model

This section defines the CubeSat model to be displayed in the simulation:

- Defining the satellite size (`sat_size`) and axis length (`axis_len`).
- Specifying the cube vertices (`v`) and faces (`f`) to represent the satellite.
- Creating an `hgtransform` object for the satellite group.
- Plotting the satellite using `patch` and setting color and edge properties.

6. Animation Loop

This is the main section that manages the simulation animation:

- Printing an animation start message.
- A `for` loop for each simulation step.
- Applying Earth rotation (`T_earth`) using `makehgtform` and `set` .
- Calculating burn progress (`burn_progress`) and applying a scale factor (`scale_factor`) to shrink the satellite during burn-up.
- Applying translation and scaling to the satellite (`T_sat`).
- Hiding the satellite when burn-up is complete (`Visible` , `off`).
- Updating the plot (`drawnow`) and pausing (`pause`) to create the animation effect.
- Printing an animation end message.